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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	E30S
Project / Solution Name	BISCOACH
Name (Leader)	MUHAMMAD HAIRIL ZAFRAN BIN MOHD HELMI
Email & Phone (Leader)	0102280047 / hairilzafran4@gmail.com
University Affiliation(s)	UNIVERSITI MALAYA
Live Prototype URL	https://biscoach.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/mirulkasy/SDC_Hackathon_E30S_Submission.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

E30S connects its components in a single end-to-end flow. Owners and staff use a mobile-first Next.js and React web app, signing in through Firebase Authentication with email and password, while barcode scanning runs directly in the browser via ZXing.

Each request passes to server-side Next.js Route Handlers under /api/*, which handle chat, receipt processing, vendor-overcharge checking, text-to-inventory and video ideas. Non-AI logic, namely the posting-schedule recommender and the pricing formula, runs in the same layer.

For AI tasks, these routes call Google Gemini (gemini-3.5-flash, with gemini-2.5-flash as fallback) through the @google/genai SDK and receive schema-constrained JSON. Multi-key failover, retries and token-credit monitoring improve resilience. In the vendor check, Gemini only reads documents; deterministic code makes the judgments.

Control stays with people. Any AI-proposed change appears as an Allow/Deny or Apply card, and nothing is saved until the user approves it. Approved changes are then written to Cloud Firestore, in organisation-scoped collections for products, transactions, tickets, checklist and publicity, with money stored as integer sen. Real-time onSnapshot listeners push updates back to the interface so owners and staff stay synchronised.

Finally, the whole application is deployed and published through Replit.

"I attach the diagram at the appendix section for the system block diagram"

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	**Purpose in Project**	**License**
JavaScript (ECMAScript) / JSX	Main programming language for frontend and server routes	Open language standard, ECMA-262
Node.js v24.21.0	Server runtime for Next.js and scheduled jobs	MIT
Next.js 16.4.2	Full-stack framework for the mobile-first UI and server-side API routes	MIT
React 19.3.0	UI library for the Transaction, Inventory, Publicity, Summary and Chatbot screens	MIT
React DOM 19.3.0	Renders React components in the browser	MIT
firebase 13.1.0	Firebase Auth sign-in, Cloud Firestore, Cloud Storage and Cloud Messaging access	Apache-2.0
@google/genai 2.30.0	Calls the Gemini API (including structured JSON)	Apache-2.0
@zxing/library 0.21.3	Barcode decoding for scanning products	MIT
@zxing/browser 0.2.1	Browser camera integration for the barcode scanner	MIT
babel-plugin-react-compiler 1.0.0	Optimizes React component rendering at build time (dev tool)	MIT
ESLint 9.39.5	Code linting and quality checks (dev tool)	MIT
eslint-config-next 14.2.35	Next.js ESLint rule set (dev tool)	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Gemini API (Google AI Studio), Google	Powers the chatbot, publicity content generation, and structured JSON outputs (ledger, summaries, reports)	free
Firebase Authentication, Google	Sign-in and account management for SME owners	free
Cloud Firestore, Google (Firebase)	Stores transactions, inventory, reports and user data	free
Cloud Storage for Firebase, Google	Stores uploaded images and generated files	free
Firebase Cloud Messaging, Google	Sends push reminders for stock and expiry alerts	free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments, Replit	Hosts the Next.js app (UI and server routes)	Managed by Replit (not user-selectable)
Replit Scheduled Deployment, Replit	Runs scheduled jobs for stock and expiry checks	Managed by Replit (not user-selectable)
Firebase Authentication, Google	User sign-in and account management	Global (Google-managed)
Cloud Firestore, Google (Firebase)	Database for transactions, inventory, reports and user data	asia-southeast1 (Singapore)
Cloud Storage for Firebase, Google	Stores uploaded images and generated files	asia-southeast1 (Singapore)

Service & Provider	Purpose in Project	Plan / Region
Firebase Cloud Messaging, Google	Delivers push reminders to users' devices	Global (Google-managed)

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 3.5 Flash (model ID: gemini-3.5-flash)	Google, accessed through the Gemini API (Google AI Studio)	pre-trained	Proprietary: Google APIs Terms of Service + Gemini API Additional Terms of Service (https://ai.google.dev/gemini-api/terms)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	**Source / URL**	**Type (public / private / self-collected / synthetic)**	**License**	**Personal Data? (Y/N)**
E-Commerce & Food Delivery Platform Fee Reference Data	Shopee MY Seller Education Hub, Lazada University, TikTok Shop Seller Center, Grab & Foodpanda Merchant Portals	Public	Public Domain / Educational Use	N
User's input (images, inventory, transaction, invoices, etc)	Entered by users in the app	Private	N/A	Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude (Claude Sonnet 5.5)	Explaining programming concepts, writing and debugging code, and adding features to our project code.	Major
Gemini API (gemini 3.5 flash)	Integrated into our project through an API key to power (feature, e.g. AI chatbot)	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Claude was used mainly to build features such as the receipt scanner, chatbot, publicity assistant and transaction history. The Gemini API powers the AI inside all of these features. Our team reviewed all generated code before adding it to the project and fixed errors and unclear logic. We tested each feature with sample inputs. As example, test receipts and invoices. We also checked that the outputs were correct and reasonable. We understand how the code works and take full responsibility for the final project, including any AI-generated parts.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit platform (cloud hosting and deployment)	Biscoach Next.js app is deployed and published through Replit, which hosts both the web interface and the backend API routes.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was very *helpful* but ***not strictly essential***; BISCOACH could be moved to another host. Its deployment and public domain made our prototype instantly accessible to judges and testers.

The core logic is a standard Next.js application, with Firebase Authentication, Cloud Firestore and Google Gemini handling sign-in, data and AI, so the codebase is portable. We did most of our coding in VS Code, using Replit mainly for deployment and publishing. Platforms such as Netlify could host the same code with minimal changes, but Replit made launching quick and straightforward

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Our application collects the user's name, company name, company type, and estimated monthly sales. This information is used to provide the application's intended functionality. Data is protected through appropriate security measures, such as access controls and secure data transmission, where implemented. Data is retained only as long as necessary for the application's purposes and is deleted when no longer needed. Any data shared with third-party services, including AI APIs, is handled according to the services' privacy policies.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The system runs on Google AI Studio's free API tier, so it can hit rate limits and quota caps during busy periods. It also relies entirely on Google's infrastructure, with no fallback if the provider goes down. The AI only acts on explicit user commands and chat history, so it depends on what the user tells it.

To prevent hallucinations and unintended actions like accidental inventory changes, the app uses a human-in-the-loop (HITL) safeguard where the AI asks for approval before updating stock or changing any data. Conversational memory across sessions also helps it track user intent and reduces factual drift.

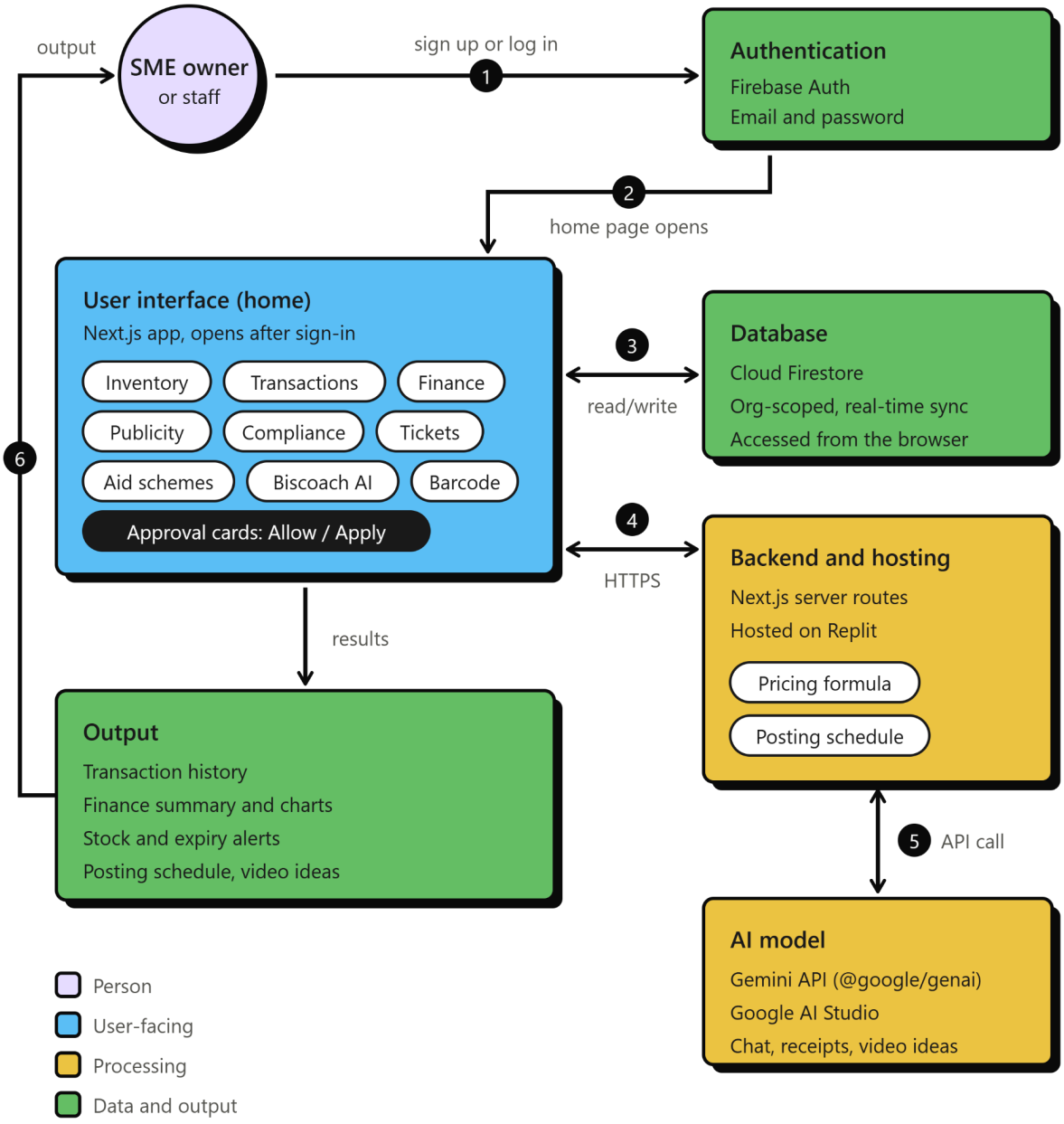
7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Appendix

2.1.



9. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



MUHAMMAD HAIRIL ZAFRAN BIN MOHD HELMI
24 OCTOBER 2026